NIST Atomic Spectra Database Ionization Energies Data

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Multiple spectra
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At. Num.	El. name	Ground Shells[a]	Ground Level	Ionization Energy (eV)	Uncertainty[b] (eV)	References
1	Hydrogen	$1s$	$^2S_{1/2}$	(13.598434599702)	0.000000000012	HDEL
2	Helium	$1s^2$	$^1S_0$	24.587389011	0.000000025	L17714
3	Lithium	$1s^2 2s$	$^2S_{1/2}$	5.391714996	0.000000022	L12261
4	Beryllium	$1s^2 2s^2$	$^1S_0$	9.322699	0.000007	L5964
5	Boron	$1s^2 2s^2 2p$	$^2P^\circ_{1/2}$	8.298019	0.000003	L12312
6	Carbon	$1s^2 2s^2 2p^2$	$^3P_0$	11.2602880	0.0000011	L20057
7	Nitrogen	$1s^2 2s^2 2p^3$	$^4S^\circ_{3/2}$	14.53413	0.00004	L1411
8	Oxygen	$1s^2 2s^2 2p^4$	$^3P_2$	13.618055	0.000007	L74,L3760
9	Fluorine	$1s^2 2s^2 2p^5$	$^2P^\circ_{3/2}$	17.42282	0.00005	L7481,L526
10	Neon	$1s^2 2s^2 2p^6$	$^1S_0$	21.564541	0.000007	L8826,L1407
11	Sodium	$[Ne]3s$	$^2S_{1/2}$	5.13907696	0.00000025	L10921,L9648
12	Magnesium	$[Ne]3s^2$	$^1S_0$	7.646236	0.000004	L10635,L6969
13	Aluminum	$[Ne]3s^2 3p$	$^2P^\circ_{1/2}$	5.985769	0.000003	L7215,L10321
14	Silicon	$[Ne]3s^2 3p^2$	$^3P_0$	8.15168	0.00003	L5815
15	Phosphorus	$[Ne]3s^2 3p^3$	$^4S^\circ_{3/2}$	10.486686	0.000015	L5148
16	Sulfur	$[Ne]3s^2 3p^4$	$^3P_2$	10.3600167	0.0000014	L23116
17	Chlorine	$[Ne]3s^2 3p^5$	$^2P^\circ_{3/2}$	12.967633	0.000016	L19191
18	Argon	$[Ne]3s^2 3p^6$	$^1S_0$	15.7596119	0.0000005	L9217
19	Potassium	$[Ar]4s$	$^2S_{1/2}$	4.34066373	0.00000009	L5451,L5783,L7185
20	Calcium	$[Ar]4s^2$	$^1S_0$	6.1131549210	0.0000000005	L22848
21	Scandium	$[Ar]3d4s^2$	$^2D_{3/2}$	6.56149	0.00006	L7185,L2110
22	Titanium	$[Ar]3d^2 4s^2$	$^3F_2$	6.828120	0.000012	L17996
23	Vanadium	$[Ar]3d^3 4s^2$	$^4F_{3/2}$	6.746187	0.000021	L17712,L20010
24	Chromium	$[Ar]3d^5 4s$	$^7S_3$	6.76651	0.00004	L7185,L2832
25	Manganese	$[Ar]3d^5 4s^2$	$^6S_{5/2}$	7.4340380	0.0000012	L19057
26	Iron	$[Ar]3d^6 4s^2$	$^5D_4$	7.9024681	0.0000012	L7743c109
27	Cobalt	$[Ar]3d^7 4s^2$	$^4F_{9/2}$	7.88101	0.00012	L10545
28	Nickel	$[Ar]3d^8 4s^2$	$^3F_4$	7.639878	0.000017	L12369
29	Copper	$[Ar]3d^{10} 4s$	$^2S_{1/2}$	7.726380	0.000004	L14897
30	Zinc	$[Ar]3d^{10} 4s^2$	$^1S_0$	9.394197	0.000006	L13951
31	Gallium	$[Ar]3d^{10} 4s^2 4p$	$^2P^\circ_{1/2}$	5.9993020	0.0000012	L5566
32	Germanium	$[Ar]3d^{10} 4s^2 4p^2$	$^3P_0$	7.899435	0.000012	L3935,L12369
33	Arsenic	$[Ar]3d^{10} 4s^2 4p^3$	$^4S^\circ_{3/2}$	9.78855	0.00025	L1351
34	Selenium	$[Ar]3d^{10} 4s^2 4p^4$	$^3P_2$	9.752368	0.000006	L21645
35	Bromine	$[Ar]3d^{10} 4s^2 4p^5$	$^2P^\circ_{3/2}$	11.81381	0.00006	L351,L67
36	Krypton	$[Ar]3d^{10} 4s^2 4p^6$	$^1S_0$	13.9996055	0.0000020	L12196
37	Rubidium	$[Kr]5s$	$^2S_{1/2}$	4.1771281	0.0000012	L4740,L5783
38	Strontium	$[Kr]5s^2$	$^1S_0$	5.69486745	0.00000012	L15350,L5644
39	Yttrium	$[Kr]4d5s^2$	$^2D_{3/2}$	6.21726	0.00010	L8844,L2111,L9735
40	Zirconium	$[Kr]4d^2 5s^2$	$^3F_2$	6.634126	0.000005	L21107
41	Niobium	$[Kr]4d^4 5s$	$^6D_{1/2}$	6.75885	0.00004	L6829
42	Molybdenum	$[Kr]4d^5 5s$	$^7S_3$	7.09243	0.00004	L6829
43	Technetium	$[Kr]4d^5 5s^2$	$^6S_{5/2}$	7.11938	0.00003	L15237
44	Ruthenium	$[Kr]4d^7 5s$	$^5F_5$	7.36050	0.00005	L10032
45	Rhodium	$[Kr]4d^8 5s$	$^4F_{9/2}$	7.45890	0.00005	L10032
46	Palladium	$[Kr]4d^{10}$	$^1S_0$	8.336839	0.000010	L20363
47	Silver	$[Kr]4d^{10} 5s$	$^2S_{1/2}$	7.576234	0.000025	L11835
48	Cadmium	$[Kr]4d^{10} 5s^2$	$^1S_0$	8.993820	0.000016	L18843
49	Indium	$[Cd]5p$	$^2P^\circ_{1/2}$	5.7863558	0.0000005	L21649
50	Tin	$[Cd]5p^2$	$^3P_0$	7.343918	0.000012	L3936
51	Antimony	$[Cd]5p^3$	$^4S^\circ_{3/2}$	8.608389	0.000012	L8844,L6684
52	Tellurium	$[Cd]5p^4$	$^3P_2$	9.009808	0.000006	L21109
53	Iodine	$[Cd]5p^5$	$^2P^\circ_{3/2}$	10.451236	0.000025	L13998,L22518
54	Xenon	$[Cd]5p^6$	$^1S_0$	12.1298437	0.0000015	L12013,L9794,L6426
55	Cesium	$[Xe]6s$	$^2S_{1/2}$	3.89390572743	0.00000000017	L19479
56	Barium	$[Xe]6s^2$	$^1S_0$	5.2116646	0.0000012	L6347,L9224
57	Lanthanum	$[Xe]5d6s^2$	$^2D_{3/2}$	5.5769	0.0006	L3974,L16228
58	Cerium	$[Xe]4f5d6s^2$	$^1G^\circ_4$	5.5386	0.0004	L4277
59	Praseodymium	$[Xe]4f^3 6s^2$	$^4I^\circ_{9/2}$	[5.4702]	0.0004	L19006
60	Neodymium	$[Xe]4f^4 6s^2$	$^5I_4$	5.52475	0.00005	L23155
61	Promethium	$[Xe]4f^5 6s^2$	$^6H^\circ_{5/2}$	5.58187	0.00004	L20992
62	Samarium	$[Xe]4f^6 6s^2$	$^7F_0$	5.643722	0.000021	L21624
63	Europium	$[Xe]4f^7 6s^2$	$^8S^\circ_{7/2}$	5.670385	0.000005	L13382
64	Gadolinium	$[Xe]4f^7 5d6s^2$	$^9D^\circ_2$	6.14980	0.00004	L11798
65	Terbium	$[Xe]4f^9 6s^2$	$^6H^\circ_{15/2}$	5.8638	0.0006	L4277
66	Dysprosium	$[Xe]4f^{10} 6s^2$	$^5I_8$	5.939061	0.000006	L22122
67	Holmium	$[Xe]4f^{11} 6s^2$	$^4I^\circ_{15/2}$	6.0215	0.0006	L4277
68	Erbium	$[Xe]4f^{12} 6s^2$	$^3H_6$	6.1077	0.0010	L4277
69	Thulium	$[Xe]4f^{13} 6s^2$	$^2F^\circ_{7/2}$	6.184402	0.000009	L22274
70	Ytterbium	$[Xe]4f^{14} 6s^2$	$^1S_0$	6.254160	0.000012	L4980
71	Lutetium	$[Xe]4f^{14} 5d6s^2$	$^2D_{3/2}$	5.425871	0.000012	L10494,L1638
72	Hafnium	$[Xe]4f^{14} 5d^2 6s^2$	$^3F_2$	6.825070	0.000012	L14027,L10033

73	Tantalum	[Xe]$4f^{14}5d^{3}6s^{2}$	$^{4}F_{3/2}$	7.549571	0.000025	L13153
74	Tungsten	[Xe]$4f^{14}5d^{4}6s^{2}$	$^{5}D_{0}$	7.86403	0.00010	L11604
75	Rhenium	[Xe]$4f^{14}5d^{5}6s^{2}$	$^{6}S_{5/2}$	7.83352	0.00011	L11604
76	Osmium	[Xe]$4f^{14}5d^{6}6s^{2}$	$^{5}D_{4}$	8.43823	0.00020	L8105
77	Iridium	[Xe]$4f^{14}5d^{7}6s^{2}$	$^{4}F_{9/2}$	8.96702	0.00022	L8105
78	Platinum	[Xe]$4f^{14}5d^{9}6s$	$^{3}D_{3}$	8.95883	0.00010	L9735,L11689
79	Gold	[Xe]$4f^{14}5d^{10}6s$	$^{2}S_{1/2}$	9.225554	0.000004	L12088
80	Mercury	[Xe]$4f^{14}5d^{10}6s^{2}$	$^{1}S_{0}$	10.437504	0.000006	L11931,L11984
81	Thallium	[Hg]$6p$	$^{2}P^{\circ}_{1/2}$	6.1082873	0.0000012	L16353
82	Lead	[Hg]$6p^{2}$	$(^{1}/_{2},^{1}/_{2})_{0}$	7.4166799	0.0000006	L11569
83	Bismuth	[Hg]$6p^{3}$	$^{4}S^{\circ}_{3/2}$	7.285516	0.000006	L10664,L6247
84	Polonium	[Hg]$6p^{4}$	$^{3}P_{2}$	8.418070	0.000004	L20887,L20886
85	Astatine	[Hg]$6p^{5}$	$^{2}P^{\circ}_{3/2}$	9.31751	0.00008	L18394
86	Radon	[Hg]$6p^{6}$	$^{1}S_{0}$	10.74850		L272,L7432
87	Francium	[Rn]$7s$	$^{2}S_{1/2}$	4.0727411	0.0000011	L10714
88	Radium	[Rn]$7s^{2}$	$^{1}S_{0}$	5.2784239	0.0000025	L4979
89	Actinium	[Rn]$6d7s^{2}$	$^{2}D_{3/2}$	5.380235	0.000012	L22136,L17893
90	Thorium	[Rn]$6d^{2}7s^{2}$	$^{3}F_{2}$	6.30670	0.00025	L15992
91	Protactinium	[Rn]$5f^{2}6d7s^{2}$	$^{4}K_{11/2}$	[5.89]	0.12	L16359
92	Uranium	[Rn]$5f^{3}6d7s^{2}$	$^{5}L^{\circ}_{6}$	6.19405	0.00006	L5494,L7765
93	Neptunium	[Rn]$5f^{4}6d7s^{2}$	$^{6}L_{11/2}$	6.265608	0.000019	L23048
94	Plutonium	[Rn]$5f^{6}7s^{2}$	$^{7}F_{0}$	6.02576	0.00025	L15992,L11260
95	Americium	[Rn]$5f^{7}7s^{2}$	$^{8}S^{\circ}_{7/2}$	5.97381	0.00025	L15992
96	Curium	[Rn]$5f^{7}6d7s^{2}$	$^{9}D^{\circ}_{2}$	5.992241	0.000020	L22423
97	Berkelium	[Rn]$5f^{9}7s^{2}$	$^{6}H^{\circ}_{15/2}$	6.19785	0.00025	L15992
98	Californium	[Rn]$5f^{10}7s^{2}$	$^{5}I_{8}$	6.281878	0.000006	L22260
99	Einsteinium	[Rn]$5f^{11}7s^{2}$	$^{4}I^{\circ}_{15/2}$	6.36840	0.00006	L22449
100	Fermium	[Rn]$5f^{12}7s^{2}$	$^{3}H_{6}$	[6.50]	0.07	L16359
101	Mendelevium	[Rn]$5f^{13}7s^{2}$	$^{2}F^{\circ}_{7/2}$	[6.58]	0.07	L16359
102	Nobelium	[Rn]$5f^{14}7s^{2}$	$^{1}S_{0}$	6.62621	0.00005	L20870
103	Lawrencium	[Rn]$5f^{14}7s^{2}7p$	$^{2}P^{\circ}_{1/2}$	4.96	0.05	L21372
104	Rutherfordium	[Rn]$5f^{14}6d^{2}7s^{2}$	$^{3}F_{2}$	(6.02)	0.04	L19691
105	Dubnium	[Rn]$5f^{14}6d^{3}7s^{2}$	$^{4}F_{3/2}$	(6.8)	0.5	L12825
106	Seaborgium	[Rn]$5f^{14}6d^{4}7s^{2}$	0	[7.8]	0.5	L7830
107	Bohrium	[Rn]$5f^{14}6d^{5}7s^{2}$	$5/2$	[7.7]	0.5	L13544
108	Hassium	[Rn]$5f^{14}6d^{6}7s^{2}$	4	[7.6]	0.5	L13544

Notes:

[a] Designations used in the ground shell lists:

[Ne] = $1s^{2}2s^{2}2p^{6}$
[Ar] = $1s^{2}2s^{2}2p^{6}3s^{2}3p^{6}$
[Kr] = $1s^{2}2s^{2}2p^{6}3s^{2}3p^{6}3d^{10}4s^{2}4p^{6}$
[Cd] = $1s^{2}2s^{2}2p^{6}3s^{2}3p^{6}3d^{10}4s^{2}4p^{6}4d^{10}5s^{2}$
[Xe] = $1s^{2}2s^{2}2p^{6}3s^{2}3p^{6}3d^{10}4s^{2}4p^{6}4d^{10}5s^{2}5p^{6}$
[Hg] = $1s^{2}2s^{2}2p^{6}3s^{2}3p^{6}3d^{10}4s^{2}4p^{6}4d^{10}5s^{2}5p^{6}4f^{14}5d^{10}6s^{2}$
[Rn] = $1s^{2}2s^{2}2p^{6}3s^{2}3p^{6}3d^{10}4s^{2}4p^{6}4d^{10}5s^{2}5p^{6}4f^{14}5d^{10}6s^{2}6p^{6}$

[b] Uncertainties of some of the listed values are unknown.

If you did not find the data you need, please inform the ASD Team.